MIL-R-83404(USAF)
1 May 1973

MILITARY SPECIFICATION
ROCKET MOTOR MANUFACTURING
ROCKET MOTOR SR110-AD-1

1. SCOPE

1.1 This specification defines the processing procedures which are required in the manufacture of the Rocket Motor SR110-AD-1. The processing procedures include liner and motor case preparation, lining the motor case, propellant mixing, grain casting, grain curing and final rocket motor finishing.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on the date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

O-T-634	9 Oct 68	Trichloroethylene, Technical Thinner, Lacquer
TT-L-32	30 Apr 69	Lacquer, Cellulose Nitrate

Military

MIL-P-116	18 Aug 67	Preservation, Methods of
MIL-P-7962C	3 Feb 72	Primer Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting
MIL-L-9931	7 Feb 72	Lot Numbering, Dating and Preparing Data Cards for Guided Missile Explosive Assemblies, Subassemblies, and Parts
MIL-C-45662	9 Feb 62	Calibration System Requirements
MIL-T-81533	26 Sep 67	Trichloroethane 1, 1, 1

FSC 1340

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MIL-L-83066(USAF)	Liner and Inhibitor Material for Rocket Motors
MIL-R-83403(USAF)	Rocket Motor SR110-AD-1
MIL-P-83405(USAF)	Propellant, Composite ADC-109

STANDARDS

Federal

FED-STD-595	2 Jan 68	Colors

Military

MIL-STD-709	30 Jun 67	Ammunition Color Coding

MANUALS

Air Force

AFM-71-4	Packaging and Handling of Dangerous Materials for Transporatation by Military Aircraft
ATA Dangerous Articles - Tariff 13	Motor Carrier's Explosives and Dangerous Articles Tariff
ICC Regulation Tariff No. 19	Transportation of Explosives and Other Dangerous Articles by Land

DRAWINGS

Air Force

70J2222	Rocket Motor SR110-AD-1
71F48868	Igniter Assembly
71D48864	Hardware Assembly
71F48865	Grain Assembly
71D48866	Liner Assembly
71C48875	Drawing Tree

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. DEFINITIONS

3.1 Not Applicable

4. GENERAL REQUIREMENTS

4.1 Chemical processing equipment. Equipment used for mixing, transferring, and handling chemical materials required for processing the rocket motor shall be free from foreign material which will adversely affect performance. Containers in which the chemical materials are weighed or processed shall be empty prior to use, except when the net weight of the material transferred is determined by subtracting tare weight or when residual material in the containers is acceptable for subsequent use.

4.2 Weighing equipment. The calibration of weighing equipment used in conjunction with this standard shall be in accordance with Specification MIL-C-45662.

4.3 Solvents for cleaning processes. Trichoroethylene per O-T-634 or 1, 1, 1-trichloroethane per MIL-T-81533 shall be used for cleaning of parts and components of the rocket motor as specified under the detail requirements.

4.4 Wiping cloths. All wiping cloths used for the final cleaning and wiping of the surfaces of parts and components prior to the application of sealants, liners, and insulating materials shall be either clean, lint-free cotton cloths, or equivalent industrial wipes.

4.5 Handling of cleaned parts and components. Surfaces of parts and components which have been cleaned and made ready for the application of sealants, liners, or insulating materials shall be protected from contaminants, and shall not be touched or handled with bare hands. When working with the cleaned surfaces, personnel shall wear clean outer apparel and clean gloves.

4.6 Preparation of sealants, liners, inhibitors and insulations. Sealants, liners, inhibitors, and insulating materials shall be prepared according to the type of material. If required, these materials shall be mixed in accordance with 4.6.1 unless otherwise specified under the detail requirements.

4.6.1 Mixing. Prior to mixing, it shall be determined that the raw materials being used conform to the applicable specifications or standards. The material shall then be mixed until the ingredients of the formulation are uniformly dispersed. The essential criteria for uniform dispersion shall be sufficient wetness of the material solids, and absence of lumps in the material.

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4.7 Processing steps. The sequence of operations specified herein may be varied as necessary and accomplished concurrently to meet schedules and to facilitate processing and manufacture.

5. DETAIL REQUIREMENTS

5.1.1 Hand degreasing. The motor case shall be cleaned as required to remove preservatives, oils, grease, discoloration, and any other contaminations. Solvent cleaning shall be accomplished by hand degreasing. Interior surfaces of the case shall be wiped with a clean lint-free cloth which has been moistened with solvent. Excess solvent shall be wiped from the surfaces with clean lint-free cloths and the surfaces allowed to dry.

5.1.2 Dimensional inspection. The unlined motor case without headcap is to be weighed. The lined motor case without headcap is to be weighed. The loaded motor case without headcap is to be weighed. Records of these weights are to be kept.

5.2 Liner processing. The ingredients and procedures shall be controlled to produce a uniform material and coating to meet the requirements of MIL-L-83066 and MIL-R-83403(USAF) Rocket Motor SR110-AD-1.

5.2.1 Liner mixing. The mixing equipment and procedure shall produce a material that is thoroughly blended and smooth flowing.

5.2.2 Liner application. The liner application shall produce a complete coating of liner meeting the requirements of 5.2.4.

5.2.3 Liner curing. The liner shall be cured at a time and temperature to produce a solid cure with surface tack to provide propellant bond. Advisory time and temperature are 180°±5° for 16 hours ± 5 hours.

5.2.4 Liner inspection. Liner shall be visually inspected for defective coverage and weight determined per paragraph 5.1.2 to assure compliance with the requirements of Drawing 71D48866.

5.3 Propellant processing. The ingredients, equipment and procedures shall be controlled to produce a uniform propellant with adequate physical properties to meet the requirements of MIL-P-83405(USAF) and ballistic properties to meet the requirements of MIL-R-83403(USAF).

5.3.1 Propellant mixing. The mixing equipment and procedure shall produce a thorough blend of the raw materials and provide smooth flowing material having a pot life of four-hours to six-hours with no degradation of ballistic properties of the finished motor.

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5.3.2 Propellant casting. The propellant casting procedures shall produce a completely filled motor case of sufficient weight to meet the requirements of 5.3.4 and with voids of lesser number and smaller size than allowed in 5.3.4.

5.3.3 Propellant curing. The propellant shall be cured at a time and temperature to produce the physical property requirements of MIL-P-83405(USAF) Propellant, Composite ADC-109. Advisory time and temperature are 180°±5° for 48 hours ± 5 hours.

5.3.4 Propellant inspection. The completed grain shall be visually inspected for complete filling, weight determined per paragraph 5.1.2, to assure compliance with Drawing 71F48865, and X-ray inspected to determine size, location and number of subsurface defects. Grain shall meet all requirements of Drawing 71F48865 and MIL-R-83403(USAF).

5.4 Rocket motor final assembly. The rocket motor final assembly shall be controlled to meet the drawing requirements with care to assure proper matching of cap and case by number and scribe mark. The headcap interface with dart surfaces shall be protected to avoid damage.

5.4.1 Headcap assembly. The headcap shall be bonded and held in place with removable pins during X-ray. Head cap shall be pinned per Drawing 71D48864 following X-ray acceptance.

5.4.2 Igniter ring bonding. Igniter mounting ring shall be bonded per Drawing 70J2222. The nozzle closure is screwed into the igniter mounting ring prior to shipment per Drawing 70J2222.

5.5 Rocket motor finishing. The assembled motor shall be painted, marked and labeled per Drawing 70J2222.

5.5.1 Painting. The motor shall be thoroughly cleaned and dryed prior to application of primer coat MIL-P-7962 and finish coat of lacquer TT-L-32.

5.5.2 Labeling. Labels shall be applied per Drawing 71C48869.

5.6 Preparation for delivery.

5.61. Preservation, packaging and packing. Prior to packing for shipment and storage, the exterior of the rocket motor shall be free from visible grease, dirt, or other contaminants. Cleaning, if required, shall be by any process or combination of processes which will accomplish thorough cleaning without damage to the item. Preservation and packaging shall comply with Method III of Specification MIL-P-116 which requires that unpreserved items be packaged for mechanical and physical protection only.

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5.6.2 Marking for shipment and storage. The marking on the shipping container shall be inspected to ensure compliance with applicable T.P.O.

5.6.2.1 Safety notice. The rocket motor is a Class B explosive. Warning labels on the shipping containers, when required, shall conform to the requirements of T.C. George Tariff No. 19, ATA Dangerous Articles Tariff 13 or AFM 71-4 as applicable to the mode of transportation.

Custodian:

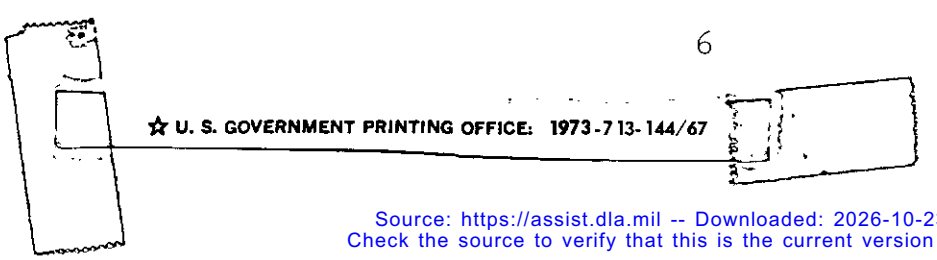
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